



1. NAME OF THE MEDICINAL PRODUCT

AVAZET 10mg/10mg Film Coated Tablets
AVAZET 10mg/20mg Film Coated Tablets
AVAZET 10mg/40mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

AVAZET 10mg/10mg Film Coated Tablets
Each tablet contains 10mg of Ezetimibe and 10.825mg of Atorvastatin Calcium Trihydrate equivalent to 10mg of Atorvastatin.

AVAZET 10mg/20mg Film Coated Tablets

Each tablet contains 10mg of Ezetimibe and 21.65mg of Atorvastatin Calcium Trihydrate equivalent to 20mg of Atorvastatin.

AVAZET 10mg/40mg Film Coated Tablets

Each tablet contains 10mg of Ezetimibe and 43.30mg of Atorvastatin Calcium Trihydrate equivalent to 40mg of Atorvastatin.

For the full list of excipients, see section List of excipients.

3. PHARMACEUTICAL FORM

Film Coated Tablets.

AVAZET 10mg/10mg Film Coated Tablets

White to off-white, capsule shaped, film-coated tablets debossed with "F5" on one side and plain on the other side.

AVAZET 10mg/20mg Film Coated Tablets

White to off-white, capsule shaped, film-coated tablets debossed with "F7" on one side and plain on the other side.

AVAZET 10mg/40mg Film Coated Tablets

White to off-white, capsule shaped, film-coated tablets debossed with "F11" on one side and plain on the other side.

4. CLINICAL PARTICULARS

4.1Therapeutic indications

Prevention of Cardiovascular Events

AVAZET is indicated to reduce the risk of cardiovascular events [cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization] in patients with coronary heart disease (CHD) and a history of acute coronary syndrome (ACS), either previously treated with a statin or not.

Hypercholesterolaemia

AVAZET is indicated as adjunctive therapy to diet for use in adults with primary (heterozygous familial and non-familial) hypercholesterolaemia or mixed hyperlipidaemia where use of a combination product is appropriate

- patients not appropriately controlled with a statin alone
- patients already treated with a statin and ezetimibe

Homozygous Familial Hypercholesterolaemia (HoFH)

AVAZET is indicated as adjunctive therapy to diet for use in adults with HoFH. Patients may also receive adjunctive treatments (e.g., low-density lipoprotein [LDL] apheresis).

4.2 Posology and method of administration

Hypercholesterolaemia and/or Coronary Heart Disease (with ACS History)

The patient should be on an appropriate lipid lowering diet and should continue on this diet during treatment with AVAZET.

The dose range of AVAZET is 10/10 mg/day through *10/80 mg/day. The typical dose is 10/10 mg once a day. The patient's low-density lipoprotein cholesterol (LDL-C) level, coronary heart disease risk status, and response to current cholesterol-lowering therapy should be considered when starting therapy or adjusting the dose.

The dose of AVAZET should be individualised based on the known efficacy of the various dose strengths of AVAZET and the response to the current cholesterol-lowering therapy. Adjustment of dose should be made at intervals of 4 weeks or more.

**AVAZET is available in strength of 10/10mg, 10/20mg and 10/40mg.*

It is not able to deliver all approved dose regime of combination Atorvastatin and Ezetimibe i.e. 10/80mg.

Other approved dosage forms and strengths of combination Atorvastatin and Ezetimibe should be used.

Homozygous Familial Hypercholesterolaemia

The dose of AVAZET in patients with homozygous FH is 10/10 to *10/80 mg daily. AVAZET may be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable.

Coadministration with bile acid sequestrants

Dosing of AVAZET should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

Geriatric patients

No dose adjustment is required for older patients.

Paediatric population

The safety and efficacy of AVAZET in children has not been established. No data are available.

Patients with hepatic impairment

AVAZET should be used with caution in patients with hepatic impairment. AVAZET is contraindicated in patients with active liver disease.

Patients with renal impairment

No dose adjustment is required for renally impaired patients.

Dosage in Patients Taking Cyclosporine, Clarithromycin, Itraconazole, or Certain Protease Inhibitors

In patients taking cyclosporine or the HIV protease inhibitors (tipranavir plus ritonavir) or the hepatitis C protease

inhibitor (telaprevir), therapy with AVAZET should be avoided.

In patients with HIV taking lopinavir plus ritonavir, caution should be used when prescribing AVAZET and the lowest dose necessary employed.

In patients taking clarithromycin, itraconazole, or in patients with HIV taking a combination of saquinavir plus ritonavir, darunavir plus ritonavir, fosamprenavir, or fosamprenavir plus ritonavir, therapy with AVAZET should be limited to 10mg/20 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of atorvastatin is employed.

In patients taking the HIV protease inhibitor nelfinavir or the hepatitis C protease inhibitor boceprevir, therapy with AVAZET should be limited to 10mg/40 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of atorvastatin is employed.

Other Concomitant Lipid-Lowering Therapy

The combination of AVAZET and fibrates is not recommended.

Method of administration

AVAZET is for oral administration. AVAZET can be administered as a single dose at any time of the day, with or without food.

4.3 Contraindications

- AVAZET is contraindicated in patients with hypersensitivity to ezetimibe, atorvastatin, or any of its inactive ingredients.
- Active liver disease or unexplained persistent elevations of serum transaminases.
- Pregnancy and nursing

4.4 Special warnings and precautions for use

Myopathy/Rhabdomyolysis.

Rare cases of rhabdomyolysis with acute renal failure secondary to myoglobinuria have been reported with atorvastatin and with other drugs in this class. A history of renal impairment may be a risk factor for the development of rhabdomyolysis. Such patients merit closer monitoring for skeletal muscle effects. Atorvastatin, like other statins, occasionally causes myopathy, defined as muscle aches or muscle weakness in conjunction with increases in CPK values >10 times ULN. Myopathy should be considered in any patient with diffuse myalgias, muscle tenderness or weakness, and/or marked elevation of CPK. Patients should be advised to report promptly unexplained muscle pain, tenderness or weakness, particularly if accompanied by malaise or fever or if muscle signs and symptoms persist after discontinuing AVAZET. AVAZET therapy should be discontinued if markedly elevated CPK levels occur or myopathy is diagnosed or suspected. The risk of myopathy during treatment with statins is increased with concurrent administration of cyclosporine, fibric acid derivatives, erythromycin, clarithromycin, the hepatitis C antiviral agents telaprevir, elbasvir, grazoprevir, combinations of HIV protease inhibitors, including saquinavir plus ritonavir, lopinavir plus ritonavir, tipranavir plus ritonavir, darunavir plus ritonavir, fosamprenavir, and fosamprenavir plus ritonavir, niacin, or azole antifungals. Physicians considering combined therapy with AVAZET and fibrates, erythromycin, immunosuppressive drugs, azole antifungals, or lipid- modifying doses of niacin (≥ 1g/day) should carefully weigh the potential benefits and risks and should carefully monitor patients for any signs or symptoms of muscle pain, tenderness, or weakness, particularly during the initial months of therapy and during any periods of upward dosage titration of either drug. Therefore, lower starting and maintenance doses of AVAZET should also be considered when taken concomitantly with the aforementioned drugs. Temporary suspension of AVAZET may be appropriate during fusidic acid therapy. Periodic CPK determinations may be considered in such situations, but there is no assurance that such monitoring will prevent the occurrence of severe myopathy.

Prescribing recommendations for interacting agents are summarized in Table 1

Table 1: Atorvastatin Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis

Interacting Agents	Prescribing Recommendations for AVAZET
Cyclosporine, HIV protease inhibitors (tipranavir plus ritonavir), hepatitis C protease inhibitor (telaprevir), gemfibrozil.	Avoid AVAZET.
Other fibrates (except fenofibrate), fusidic acid.	Not recommended with AVAZET.
HIV protease inhibitor (lopinavir plus ritonavir).	Use with caution and lowest dose necessary.
Clarithromycin, itraconazole, HIV protease inhibitors (saquinavir plus ritonavir*, darunavir plus ritonavir, fosamprenavir, fosamprenavir plus ritonavir), hepatitis C antiviral agents (boceprevir, elbasvir, grazoprevir).	Do not exceed 10/20 mg AVAZET daily.
HIV protease inhibitor (nelfinavir).	Do not exceed 10/40 mg AVAZET daily.

* Use with caution and with the lowest dose necessary

Cases of myopathy, including rhabdomyolysis, have been reported with atorvastatin coadministered with colchicine, and caution should be exercised when prescribing AVAZET with colchicine. Reports of myopathy and/or rhabdomyolysis have been observed with HMG-CoA reductase inhibitors coadministered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to suspending AVAZET temporarily in patients taking daptomycin.

AVAZET therapy should be temporarily withheld or discontinued in any patient with an acute, serious condition suggestive of a myopathy or having a risk factor predisposing to the development of renal failure secondary to rhabdomyolysis (e.g., severe acute infection, hypotension, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders, and uncontrolled seizures).

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation;
- improvement with immunosuppressive agents.

Myasthenia Gravis/Ocular Myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. AVAZET should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

Liver Enzymes.

In controlled coadministration trials in patients receiving ezetimibe with atorvastatin, consecutive transaminase elevations (≥ 3 X the upper limit of normal [ULN]) have been observed.

Atorvastatin, like some other lipid-lowering therapies, has been associated with biochemical abnormalities of liver function.

It is recommended that liver enzyme tests be obtained prior to initiating therapy with AVAZET and repeated as clinically indicated. There have been rare postmarketing reports of fatal and non-fatal hepatic failure in patients taking statins, including atorvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinemia or jaundice occurs during treatment with AVAZET, promptly interrupt therapy. If an alternate etiology is not found, do not restart AVAZET.

AVAZET should be used with caution in patients who consume substantial quantities of alcohol and/or have a history of liver disease. Active liver disease or unexplained persistent transaminase elevations are contraindications to the use of atorvastatin.

Endocrine Function.

Statins interfere with cholesterol synthesis and theoretically might blunt adrenal and/or gonadal steroid production. Clinical studies have shown that atorvastatin does not reduce basal plasma cortisol concentration or impair adrenal reserve. The effects of statins on male fertility have not been studied in adequate numbers of patients. The effects, if any, on the pituitary-gonadal axis in premenopausal women are unknown. Caution should be exercised if AVAZET is administered concomitantly with drugs that may decrease the levels or activity of endogenous steroid hormones, such as ketoconazole, spironolactone, and cimetidine.

Hepatic Impairment

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic impairment, AVAZET is not recommended in these patients.

Fibrates.

Gemfibrozil: Concomitant administration of AVAZET with gemfibrozil should be avoided.

Fenofibrate: Caution should be used when prescribing AVAZET and fenofibrate, as fenofibrate can cause myopathy when give alone.

If cholelithiasis is suspected in a patient receiving AVAZET and fenofibrate, gallbladder studies are indicated and alternative lipid-lowering therapy should be considered [see the product labeling for fenofibrate and fenofibric acid].

Other fibrates: The coadministration of ezetimibe with other fibrates has not been studied. Therefore, coadministration of AVAZET and other fibrates is not recommended.

Fusidic acid

Patients on fusidic acid treated concomitantly with AVAZET may have an increased risk of myopathy/rhabdomyolysis. Coadministration with fusidic acid is not recommended. In patients where the use of systemic fusidic acid is considered essential, AVAZET should be discontinued throughout the duration of fusidic acid treatment. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for coadministration of AVAZET and fusidic acid should only be considered on a case-by-case basis under close medical supervision.

Anticoagulants

If AVAZET is added to warfarin, another coumarin anticoagulant, or flutidione, the International Normalized Ratio (INR) should be appropriately monitored.

Use in Patients with Recent Stroke or TIA

In a post-hoc analysis of the Stroke Prevention by Aggressive Reduction in Cholesterol Levels (SPARCL) study where atorvastatin 80 mg vs. placebo was administered in 4,731 subjects without CHD who had a stroke or TIA within the preceding 6 months, a higher incidence of hemorrhagic stroke was seen in the atorvastatin 80 mg group compared to placebo. The incidence of fatal hemorrhagic stroke was similar across treatment groups. The incidence of nonfatal hemorrhagic stroke was significantly higher in the atorvastatin group as compared to the placebo group. Some baseline characteristics, including hemorrhagic and lacunar stroke on study entry, were associated with a higher incidence of hemorrhagic stroke in the atorvastatin group.

Skeletal Muscle Effects

Physicians considering combined therapy with atorvastatin and fibrates, erythromycin, immunosuppressive drugs, azole antifungals, or lipid-modifying doses of niacin (≥1g/day) should carefully weigh the potential benefits and risks and should carefully monitor patients for any signs and symptoms of muscle pain, tenderness, or weakness, particularly during the initial months of therapy and during any periods of upward dosage titration of either drug. Therefore, lower starting and maintenance doses of atorvastatin should also be considered when taken concomitantly with the aforementioned drugs. Temporary suspension of atorvastatin may be appropriate during fusidic acid therapy.

4.5 Interaction with other medicinal products and other forms of interaction

AVAZET

No clinically significant pharmacokinetic interaction was seen when ezetimibe was coadministered with atorvastatin.

Multiple mechanisms may contribute to potential interactions with HMG Co-A reductase inhibitors. Drugs or herbal products that inhibit certain enzymes (e.g. CYP3A4) and/or transporter (e.g. OATP1B) pathways may increase atorvastatin plasma concentrations and may lead to an increased risk of myopathy/rhabdomyolysis.

Consult the prescribing information of all concomitantly used drugs to obtain further information about their potential interactions with atorvastatin and/or the potential for enzyme or transporter alterations and possible adjustments to dose and regimens.

CYP3A4 Interactions.

Ezetimibe does not induce cytochrome P450 drug metabolizing enzymes. No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolized by cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase. Atorvastatin is metabolized by cytochromes P450 3A4. Concomitant administration of atorvastatin with inhibitors of cytochrome P450 3A4 can lead to increases in plasma concentrations of atorvastatin. The extent of interaction and potentiation of effects depends on the variability of effect on cytochrome P450 3A4. Inhibitors of cytochrome P3A4 increase the risk of myopathy by reducing the elimination of the atorvastatin component of AVAZET:

- Clarithromycin:** Atorvastatin AUC was significantly increased with concomitant administration of 80 mg atorvastatin with clarithromycin (500 mg twice daily) compared to that of atorvastatin alone. Therefore, in patients taking clarithromycin, caution should be used when the AVAZET dose exceeds 10/20 mg
- Combination of Protease Inhibitors:** Atorvastatin AUC was significantly increased with concomitant administration of atorvastatin with several combinations of HIV protease inhibitors, as well as with the hepatitis C protease inhibitor telaprevir, compared to that of atorvastatin alone. Therefore, in patients taking the HIV protease inhibitor tipranavir plus ritonavir, or the hepatitis C protease inhibitor telaprevir, concomitant use of AVAZET should be avoided. In patients taking the HIV protease inhibitor lopinavir plus ritonavir, caution should be used when prescribing AVAZET, and the lowest dose necessary should be used. In patients taking the HIV protease inhibitors saquinavir plus ritonavir, darunavir plus ritonavir, fosamprenavir, or fosamprenavir plus ritonavir, or the hepatitis C protease inhibitor boceprevir, the dose of AVAZET should not exceed 10/20 mg and should be used with caution. In patients taking the HIV protease inhibitor nelfinavir the dose of AVAZET should not exceed 10/40 mg and close clinical monitoring is recommended.
- Itraconazole:** Atorvastatin AUC was significantly increased with concomitant administration of atorvastatin 40 mg and itraconazole 200 mg. Therefore, in patients taking itraconazole, caution should be used when the AVAZET dose exceeds 10/20 mg.
- Grapefruit Juice:** Contains one or more components that inhibit CYP 3A4 and can increase plasma concentrations of atorvastatin, especially with excessive grapefruit juice consumption (>1.2 liters per day).

- Cyclosporine:** Atorvastatin and atorvastatin-metabolites are substrates of the OATP1B1 transporter. Inhibitors of the OATP1B1 (e.g., cyclosporine) can increase the bioavailability of atorvastatin. Atorvastatin AUC was significantly increased with concomitant administration of atorvastatin 10 mg and cyclosporine 5.2 mg/kg/day compared to that of atorvastatin alone. The coadministration of AVAZET with cyclosporine should be avoided.

Other Interactions

Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant. Coadministration of atorvastatin with an oral antacid suspension containing magnesium and aluminium hydroxides decreased plasma concentrations of atorvastatin and its active metabolites approximately 35%; however, LDL-C reduction was not altered.

Cholestyramine: Concomitant cholestyramine administration decreased the mean AUC of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55%. The incremental LDL-C reduction due to adding ezetimibe to cholestyramine may be lessened by this interaction.

Fibrates:

- Gemfibrozil:** Due to an increased risk of myopathy/rhabdomyolysis when HMG-CoA reductase inhibitors are coadministered with gemfibrozil, concomitant administration of AVAZET with gemfibrozil should be avoided.
- Fenofibrate:** Because it is known that the risk of myopathy during treatment with HMG-CoA reductase inhibitors is increased with concurrent administration of fenofibrate, AVAZET should be administered with caution when used concomitantly with fenofibrate.
- Other fibrates:** The safety and effectiveness of ezetimibe administered with other fibrates have not been established. Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. Coadministration of AVAZET with other fibrates is not recommended until use in patients is studied.

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Fusidic Acid: The risk of myopathy/rhabdomyolysis may be increased by concomitant administration of fusidic acid.

Anticoagulants: Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time in a study of twelve healthy adult males. There have been post-marketing reports of increased INR in patients who had ezetimibe added to warfarin or flutidione. Most of these patients were also on other medications. Atorvastatin had no clinically significant effect on prothrombin time when administered to patients receiving chronic warfarin treatment. The effect of AVAZET on the prothrombin time has not been studied.

Inhibitors of Breast Cancer Resistant Protein (BCRP): Atorvastatin is a substrate of the efflux transporter BCRP. Concomitant administration of products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of atorvastatin and an increased risk of myopathy; therefore, a dose adjustment of atorvastatin may be necessary. Coadministration of elbasvir and grazoprevir with atorvastatin increases plasma concentrations of atorvastatin by 1.9-fold due in part to CYP3A and/or BCRP inhibition; therefore, the dose of AVAZET should not exceed 10/20 mg daily in patients receiving concomitant medication with products containing elbasvir or grazoprevir.

Inducers of Cytochrome P450 3A4: Concomitant administration of atorvastatin with inducers of cytochrome P450 3A4 (e.g., efavirenz, rifampin) can lead to variable reductions in plasma concentrations of atorvastatin. Due to the dual interaction mechanism of rifampin, simultaneous coadministration of atorvastatin with rifampin is recommended, as delayed administration of atorvastatin after administration of rifampin has been associated with a significant reduction in atorvastatin plasma concentrations.

Antipyrine: Because atorvastatin does not affect the pharmacokinetics of antipyrine, interactions with other drugs metabolized via the same cytochrome isozymes are not expected.

Colestipol: Plasma concentrations of atorvastatin decreased approximately 25% when colestipol and atorvastatin were coadministered. However, LDL-C reduction was greater when atorvastatin and colestipol were coadministered than when either drug was given alone.

Digoxin: When multiple doses of atorvastatin and digoxin were coadministered, steady-state plasma digoxin concentrations increased by approximately 20%. Patients taking digoxin should be monitored appropriately.

Oral Contraceptives: Coadministration of atorvastatin and an oral contraceptive increased AUC values for norethindrone and ethinyl estradiol by approximately 30% and 20%. These increases should be considered when selecting an oral contraceptive for a woman taking atorvastatin.

Amlodipine: In a drug-drug interaction study in healthy subjects, coadministration of atorvastatin 80 mg and amlodipine 10 mg resulted in an 18% increase in exposure to atorvastatin that was not clinically meaningful.

Niacin: The risk of skeletal muscle effects may be enhanced when AVAZET is used in combination with niacin; a reduction in AVAZET dosage should be considered in this setting.

Colchicine: Cases of myopathy, including rhabdomyolysis, have been reported with atorvastatin coadministered with colchicine, and caution should be exercised when prescribing AVAZET with colchicine.

Daptomycin: The risk of myopathy and/or rhabdomyolysis may be increased by concomitant administration of HMG-CoA reductase inhibitors and daptomycin.

4.6 Fertility, pregnancy and lactation

Pregnancy

Atherosclerosis is a chronic process and discontinuation of lipid-lowering drugs during pregnancy should have little impact on the outcome of long-term therapy of primary hypercholesterolemia.

AVAZET

AVAZET is contraindicated during pregnancy. Because HMG-CoA reductase inhibitors decrease cholesterol synthesis and possibly the synthesis of other biologically active substances derived from cholesterol, AVAZET may cause fetal harm when administered to pregnant women. AVAZET should be discontinued as soon as pregnancy is recognized. Advise females of reproductive potential to use effective contraception during treatment of AVAZET.

Ezetimibe

No clinical data on exposed pregnancies are available.

Atorvastatin

Limited published data on atorvastatin from observational studies, meta-analyses and case reports have not shown an increased risk of major congenital malformations or miscarriage. Rare reports of congenital anomalies have been received following intrauterine exposure to other HMG-CoA reductase inhibitors. In a review of approximately 100 prospectively followed pregnancies in women exposed to simvastatin or lovastatin, the

incidences of congenital anomalies, spontaneous abortions, and fetal deaths/stillbirths did not exceed what would be expected in the general population. The number of cases is adequate to exclude a greater than or equal to three-to-four-fold increase in congenital anomalies over background incidence. In 89% of the prospectively followed pregnancies, drug treatment was initiated prior to pregnancy and was discontinued at some point in the first trimester when pregnancy was identified.

Breastfeeding

AVAZET
AVAZET is contraindicated in nursing mothers. Because of the potential for serious adverse reactions in a breastfed infant, women who are nursing should not take AVAZET.

Ezetimibe

It is not known whether ezetimibe is excreted into human breast milk.

Atorvastatin

There is no available information on the effects of atorvastatin on the breastfed infant or the effects of atorvastatin on milk production. It is not known whether atorvastatin is present in human milk.

4.7 Effects on ability to drive and use machines

No studies of the effects on the ability to drive and use of machines have been performed. However, certain side effects that have been reported with AVAZET may affect some patients' ability to drive or operate machinery. Individual responses to AVAZET may vary.

4.8 Undesirable effects

The following common or uncommon drug-related adverse experiences were reported in patients taking AVAZET:

	Common	Uncommon	Not Known
Infections and Infestations		Influenza	
Psychiatric disorders		Depression Insomnia Sleep Disorder	
Nervous system disorders		Dizziness Dysgeusia Headache Paresthesia	Myasthenia gravis
Cardiac disorders		Sinus bradycardia	
Vascular disorders		Hot flush	
Respiratory, thoracic and mediastinal disorders		Dyspnea	
Gastrointestinal disorders	Diarrhea	Abdominal discomfort Abdominal distension Abdominal pain Abdominal pain lower Abdominal pain upper Constipation Dyspepsia Flatulence Frequent bowel movements Gastritis Nausea Stomach discomfort	
Skin and subcutaneous tissue disorders		Acne Urticaria	
Musculoskeletal and connective tissue disorders	Myalgia	Arthralgia Back pain Muscle fatigue Muscle spasms Muscular weakness Pain in extremity	
General disorders and administration site conditions		Asthenia Fatigue Malaise Edema	
Investigations		ALT and/or AST increased Alkaline phosphatase increased Blood CK increased Gamma-glutamyltransferase increased Hepatic enzyme increased Liver function test abnormal Weight increased	
Eye Disorders			Ocular myasthenia

Post-marketing Experience

The following additional adverse reactions have been reported in post-marketing use with AVAZET or post-marketing use with ezetimibe or atorvastatin:

Infections and infestations	Nasopharyngitis
Blood and lymphatic system disorders	Thrombocytopenia
Immune system disorders	Hypersensitivity reactions, including anaphylaxis, angioedema, rash, and urticaria
Metabolism and nutrition disorders	Decreased appetite; anorexia; hyperglycemia; hypoglycemia
Psychiatric disorders	Nightmares
Nervous system disorders	Hypesthesia; peripheral neuropathy; myasthenia gravis
Eye disorders	Vision blurred; visual disturbance; ocular myasthenia
Ear and labyrinth disorders	Tinnitus; hearing loss
Vascular disorders	Hypertension
Respiratory, thoracic, and mediastinal disorders	Cough; pharyngolaryngeal pain; epistaxis
Gastrointestinal disorders	Dry mouth; pancreatitis; gastroesophageal reflux disease; eructation; vomiting
Hepatobiliary disorders	Hepatitis; cholelithiasis; cholecystitis; cholestasis
Skin and subcutaneous tissue disorders	Alopecia; pruritus; skin rash; erythema multiforme; angioneurotic oedema; dermatitis bullous including erythema multiforme; Stevens-Johnson syndrome and toxic epidermal necrolysis
Musculoskeletal and connective tissue disorders	Myopathy/rhabdomyolysis; neck pain; joint swelling; myositis tendonopathy, sometimes complicated by rupture, Immune-mediated necrotizing myopathy
Reproductive system and breast disorders	Gynecomastia
General disorders and administration site conditions	Chest pain; pain; edema peripheral; pyrexia
Investigations	White blood cells urine positive

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks). Increases in HbA1c and fasting serum glucose levels have been reported with HMG-CoA reductase inhibitors, including atorvastatin. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Musculoskeletal disorders:

Frequency not known: Immune-mediated necrotizing myopathy

The following adverse events have been reported with some statins:

- sexual dysfunction
- depression
- exceptional cases of interstitial lung disease, especially with long term therapy
- diabetes mellitus: frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI>30kg/m², raised triglycerides, history of hypertension)

4.9 Overdose

AVAZET

No specific treatment of overdose with AVAZET can be recommended. In the event of an overdose, symptomatic and supportive measures should be employed.

Ezetimibe

In clinical studies, administration of ezetimibe, 50 mg/day to 15 healthy subjects for up to 14 days, 40 mg/day to 18 patients with primary hyperlipidemia for up to 56 days, and 40 mg/day to 27 patients with homozygous sitosterolemia for 26 weeks, was generally well tolerated. A few cases of overdose have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious.

Atorvastatin

Due to extensive drug binding to plasma proteins, hemodialysis is not expected to significantly enhance atorvastatin clearance.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Lipid modifying agents, combinations of various lipid modifying agents, ATC code: C10BA05

Mechanism of action

AVAZET

Plasma cholesterol is derived from intestinal absorption and endogenous synthesis. AVAZET contains ezetimibe and atorvastatin, two lipid-lowering compounds with complementary mechanisms of action. AVAZET reduces elevated total-C, LDL-C, Apo B, TG, and non-HDL-C, and increases HDL-C through dual inhibition of cholesterol absorption and synthesis.

Ezetimibe

Ezetimibe inhibits the intestinal absorption of cholesterol. Ezetimibe is orally active and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g., statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols. Ezetimibe localizes at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms provide complementary cholesterol reduction.

Atorvastatin

Atorvastatin is a selective, competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methyl-glutaryl-coenzyme A to mevalonate, a precursor of sterols, including cholesterol. In animal models, atorvastatin lowers plasma cholesterol and lipoprotein levels by inhibiting HMG- CoA

reductase and cholesterol synthesis in the liver and by increasing the number of hepatic LDL receptors on the cell surface to enhance uptake and catabolism of LDL; atorvastatin also reduces LDL production and the number of LDL particles.

5.2 Pharmacokinetic properties

General Introduction

AVAZET

AVAZET has been shown to be bioequivalent to coadministration of corresponding doses of ezetimibe and atorvastatin tablets.

Absorption

AVAZET

The effects of a high-fat meal on the pharmacokinetics of ezetimibe and atorvastatin when administered as AVAZET tablets are comparable to those reported for the individual tablets.

Ezetimibe

After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe-glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection. Concomitant food administration (high fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as ezetimibe 10-mg tablets.

Atorvastatin

Atorvastatin is rapidly absorbed after oral administration; maximum plasma concentrations (C_{max}) occur within 1 to 2 hours. Extent of absorption increases in proportion to atorvastatin dose. After oral administration, atorvastatin film-coated tablets are 95% to 99% bioavailable compared to the oral solution. The absolute bioavailability of atorvastatin is approximately 12% and the systemic availability of HMG-CoA reductase inhibitory activity is approximately 30%. The low systemic availability is attributed to presystemic clearance in gastrointestinal mucosa and/or hepatic first-pass metabolism.

Distribution

Ezetimibe

Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Atorvastatin

Mean volume of distribution of atorvastatin is approximately 381 l. Atorvastatin is ≥ 98% bound to plasma proteins.

Metabolism

Ezetimibe

Ezetimibe is metabolized primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug- derived compounds detected in plasma, constituting approximately 10 to 20 % and 80 to 90 % of the total drug in plasma, respectively. Both ezetimibe and ezetimibe-glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Atorvastatin

Atorvastatin is metabolized by cytochrome P450 3A4 to ortho- and parahydroxylated derivatives and various beta-oxidation products. Apart from other pathways these products are further metabolized via glucuronidation. *In vitro*, inhibition of HMG-CoA reductase by ortho- and parahydroxylated metabolites is equivalent to that of atorvastatin. Approximately 70% of circulating inhibitory activity for HMG-CoA reductase is attributed to active metabolites.

Elimination

Ezetimibe

Following oral administration of ¹⁴C-ezetimibe (20 mg) to human subjects, total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the feces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Atorvastatin

Atorvastatin is eliminated primarily in bile following hepatic and/or extrahepatic metabolism. However, the medicinal product does not appear to undergo significant enterohepatic recirculation. Mean plasma elimination half-life of atorvastatin in humans is approximately 14 hours. The half-life of inhibitory activity for HMG-CoA reductase is approximately 20 to 30 hours due to the contribution of active metabolites.

Special Populations

Renal Impairment

Ezetimibe

After a single 10-mg dose of ezetimibe in patients with severe renal disease (mean CrCl ≤ 30 ml/min/1.73 m²), the mean AUC for total ezetimibe was increased approximately 1.5-fold, compared to healthy subjects. An additional patient in this study (post-renal transplant and receiving multiple medications, including cyclosporine) had a 12-fold greater exposure to total ezetimibe.

Atorvastatin

Renal disease has no influence on the plasma concentrations or lipid effects of atorvastatin and its active metabolites.

Hepatic Impairment

Ezetimibe

After a single 10-mg dose of ezetimibe, the mean area under the curve (AUC) for total ezetimibe was increased approximately 1.7-fold in patients with mild hepatic impairment (Child-Pugh score 5 or 6), compared to healthy subjects. In a 14-day, multiple-dose study (10mg daily) in patients with moderate hepatic impairment (Child-Pugh score 7 to 9), the mean AUC for total ezetimibe was increased approximately 4-fold on Day 1 and Day 14 compared to healthy subjects. No dosage adjustment is necessary for patients with mild hepatic impairment. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child-Pugh score >9) hepatic impairment, ezetimibe is not recommended in these patients.

Atorvastatin

Plasma concentrations of atorvastatin and its active metabolites are markedly increased (approx. 16-fold in C_{max} and approx. 11-fold in AUC) in patients with chronic alcoholic liver disease (Child-Pugh B).

Pediatric

Ezetimibe

The absorption and metabolism of ezetimibe are similar between children and adolescents (10 to 18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in the pediatric population < 10 years of age are not available.

Atorvastatin

Apparent oral clearance of atorvastatin in pediatric subjects appeared similar to that of adults when scaled allometrically by body weight as the body weight was the only significant covariate in atorvastatin population PK model with data including pediatric HeFH patients (ages 10 years to 17 years of age) in a study.

Geriatric

Ezetimibe

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥ 65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe.

Atorvastatin

Plasma concentrations of atorvastatin and its active metabolites are higher in healthy elderly subjects than in young adults while the lipid effects were comparable to those seen in younger patient populations.

Race

Based on a meta-analysis of pharmacokinetic studies with ezetimibe, there were no pharmacokinetic differences between Blacks and Caucasians.

Gender

Ezetimibe

Plasma concentrations for total ezetimibe are slightly higher (<20 %) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe.

Atorvastatin

Concentrations of atorvastatin and its active metabolites in women differ from those in men (women: approx. 20% higher for C_{max} and approx. 10% lower for AUC). These differences were of no clinical significance, resulting in no clinically significant differences in lipid effects among men and women.

Hemodialysis

Atorvastatin

While studies have not been conducted in patients with end-stage renal disease, hemodialysis is not expected to significantly enhance clearance of atorvastatin since the drug is extensively bound to plasma proteins.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Tablet core:

Calcium Carbonate, Lactose Monohydrate, Microcrystalline Cellulose, Croscarmellose Sodium, Hydroxypropyl Cellulose, Polysorbate 80, Colloidal Silicon Dioxide, Magnesium Stearate, Ferric Oxide, Sodium Lauryl Sulfate, Povidone

Film-coat:

Opadry Complete Film Coating System YS-1-7040 White

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please refer to the expiry date on the product labels.

6.4 Special precautions for storage

Store below 30°C. Store in the original package.

6.5 Nature and contents of container

Alu/Alu blisters in carton of 1 x 10's, 3 x 10's, 6 x 10's, 10 x 10's, 50 x 10's, 100 x 10's. Not all packs sizes may be marketed.

6.6 Special precautions for disposal and other handling

No special requirements.

7. MANUFACTURER

Manufactured by:

Novugen Pharma Sdn. Bhd.
No. 27, Jalan Lengkok Teknologi 2,
Taman Teknologi Enstek Fasa 1,
71760 Bandar Baru Enstek,
Negeri Sembilan, Malaysia

Product Registration Holder:

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8. DATE OF REVISION

Sep 2025